Combining Natural Language Generation and Graph Algorithms to Explain Causal Maps through Meaningful Paragraphs

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Abstract

Participatory modeling (PM) externalizes the knowledge of participants involved in managing complex socio-environmental problems. This interdisciplinary field builds on interrelated areas within Artificial Intelligence, including knowledge representation, knowledge synthesis, or knowledge elicitation. PM seeks to support diverse stakeholders in finding shared perspectives to solve problems. At first, the tools in PM are well-aligned with this objective, since they offer a transparent elicitation of knowledge by using easily interpretable formats such as causal maps. However, as individual causal maps are combined, the aggregate can become extremely intricate and opaque. PM only recently started to embrace technical advances in explainability by using Natural Language Generation (NLG) to create narratives, as a more accessible view of group-level knowledge than a large map. However, methods were partially destructive since parts of a model were lost during this translation, which could mislead participants in thinking that the model was incorrect or lacking. In addition, methods have focused on translating elements of a model into sentences, but participants expect a report formed around coherent paragraphs with appropriate transitions. In this paper, we design, implement, and evaluate a method that automatically translates causal maps into narratives without model loss and based on paragraphs. Through a case study, we compare 24 overlapping community detection algorithms (to decompose a map into paragraphs) and 5 versions of GPT (to produce paragraphs). The right combination produces text with a readability level comparable to human-written paragraphs. Our tool is open source, to support modeling groups in adapting our process to their stakeholders.

Keywords: Automatic Explanations, Conceptual Model, GPT, Gemini, Llama, Natural Language Generation, Network Science

1 Introduction

Complex social problems are often multidisciplinary and multilevel [1, 2]. For example, problems such as suicide involve a variety of stakeholders ranging from subject-matter experts (e.g., social epidemiology, clinical psychology), individuals with specific experiences (e.g., who survived an attempt or have inflicted self-harm), and the broader community (e.g., family members) [3]. Stakeholder groups have commonality but also significant divergence [4], for example as parents describe assessment procedures as rigid while staff are in favor of standardized protocols [5]. Working towards a shared solution thus requires an understanding of the *views held by each participant*. Common approaches (e.g., focus group methodology) include identifying themes within and across participant groups [6], and listing or ranking [7]. These may be followed by the assessment of potential concerns, for instance to assess how adolescents and parents *feel* with regard to sharing family pictures online [8]. While such approaches are useful to manage stakeholder groups, they do not fully capture the *mental model* of each individual with regard to the problem. Indeed, an elicitation process that only accounts for concepts but omits relationships is at a low level of systems thinking [9] hence an agreement between stakeholders may be superficial (Figure 1). Individuals may see the same facets of the problem (e.g., shared key constructs) but they would disagree on interventions as they expect different effects (e.g., different causal implications).

The practice of *participatory modeling* [10] thus elicits the mental models of individuals and analyzes their views about a system, for example in the form of a cognitive or causal map (Figure 1). Such maps externalize individual perspectives regarding the structure and function of a system, which allows analysts to find leverage points and perform trade-off analyses [11]. Participatory modeling is firmly rooted in Artificial Intelligence, as its core theme revolves around knowledge representation and reasoning. Some of the main tasks include the choice of a knowledge representation [12], the determination of a replicable protocol to elicit knowledge from individuals in a target representation [13, 14], the synthesis of individual representations to form a stakeholder group based either on pre-defined categories [15, 16] or similarities in knowledge [17], and the comparison of knowledge bases at the individual [18] or group level [19]. *Transparency* is essential across these tasks since participants act both as data providers and consumers of the ensuing analysis. Without transparency, we lose some of the expected benefits of systems building activities, such as learning among participants, fostering engagement, or buy-in from key actors [20]. Transparency is necessarily a trade-off, as summarized by Mendoza and Prabhu: "for modeling to be able to offer genuine analytical support for participatory management, it must be simple and transparent for local stakeholders and at the same time it must of be of sufficient rigor" [21].

Several algorithms and their implementation as software solutions readily support transparency, albeit only to a limited extent and for specific tasks. Maps are commonplace in educational technology, where they are created by students (e.g., using Coggle, cMap or jMAP), analyzed with respect to specific quality metrics (e.g., by HIMATT), or compared to the 'expert map' of an instructor in terms of structure and/or function [22–24]. In environmental modelling, a variety of software also allow participants to perform scenarios on their maps to test the implications of their knowledge base

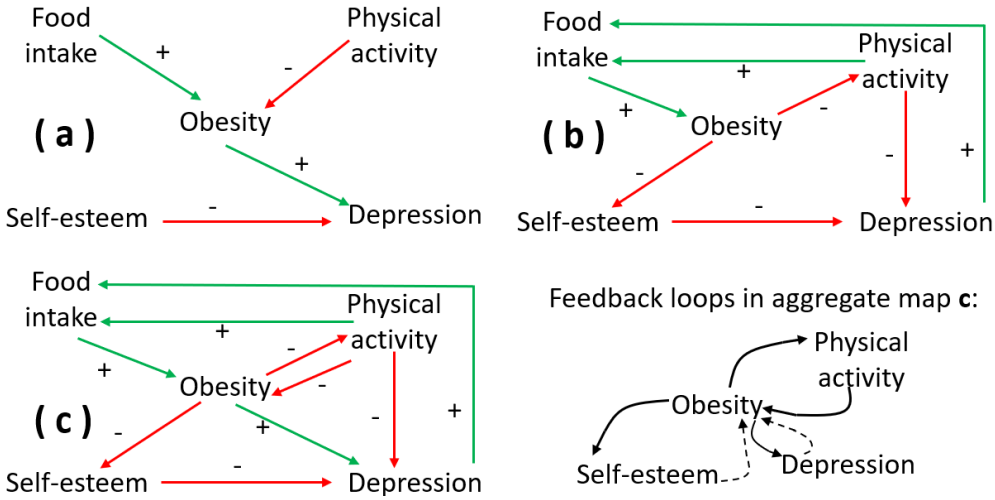

Fig. 1 Two participants working on obesity may have the same constructs and agree that physical activity and food intake are most important (a, b). However, their perceived causal relationships may be different, hence they would advocate for different interventions. When their views are aggregated (c), the system map is more comprehensive but its increased density can also raise complexity, for example through new direct and indirect feedback loops.

(e.g., using MentalModeler) or to specify how the views of participants can influence each other [25–27]. Despite the plethora of software, little support has been provided for one of the most challenging issues in transparency for participatory modelling: understanding the *aggregated* map of a group.

While each participant can navigate their own map, the map of a collective may reach a much larger number of nodes and edges, and engender several structures that are important for the dynamics of a problem (e.g., feedback loops) yet difficult to comprehend. As a result, there have been recurrent criticisms of the opacity of larger maps. A case in point is the Foresight Obesity Map, which is now portrayed as looking like a "spilled plate of spaghetti" [28–30] rather than examined for its intrinsic scientific value. The development of specialized software to interact with large systems maps has not fully solved the problem [31], as participants may be unfamiliar with navigating large diagrammatic representations of a system. Recently, we proposed to transform the maps into text, which is a familiar format for a broader audience [32]. Our proof of feasibility used Natural Language Generation via GPT-3 to create sentences from a causal map. However, it had two important limitations. First, since a text is linear but maps can have loops, our prototype decomposed a map in a way that only preserved some of the content (e.g., several edges are eliminated). Second, our solution addressed the first step of creating *sentences*, but they still need to be bundled into logical *paragraphs* for participants.

The main contribution of this paper is to design, implement, and evaluate an AI system that supports transparency in participatory modeling by transforming maps into paragraphs that are sufficiently readable for our target audience. This contribution builds on our prototype presented as a conference paper [32], and specifically addresses its two limitations by (i) decomposing a map into components that preserve its entire

content, and most importantly (ii) turning these components into topical paragraphs that are sufficiently readable.

The remainder of this paper is organized as follows. In Section 2, we provide a succinct background on the matter of Natural Language Generation for graph-to-text, along with a summary of our prototype. The methodological improvements of this article are detailed in Section 3, including our 24 graph algorithms to navigate and decompose a system and our use of GPT to produce paragraphs.The section also covers how 19 of the 24 algorithms had hyper-parameters, which we calibrated via a grid search. Results are evaluated across multiple versions of GPT (3, 3.5, 4) along with Gemini 1.0 (Gemini-1.0-pro) and Llama-2 (Llama-2-7b-chat-hf) in Section 4 and contextualized in Section 5 with respect to advancing the application of AI methods in participatory modeling.

2 Background

2.1 Core Steps in Graph-to-text Generation

Our work participates to the broader domain of *data-to-text generation*, which consists of producing a textual description to help readers in understanding non-linguistic data (e.g., tables, numbers, graphs). Within this domain, we focus on the subtask of *graph-to-text generation* [33], in which the input data consists of a set of nodes and edges. Since the input could be a large graph with cycles but text generators may expect a sequence of tokens, the graph undergoes a *linearization step* that decomposes it into a suitable format, such as a triplet [source node, relation, target node] for every edge. Although linearization can be obtained by a graph traversal, neighboring nodes in the original graph may become far away in the final serialization, hence various processes modify or augment graph traversals to preserve structural information [34].

The input graph in the literature is often a knowledge graph [35], where nodes and edges are labeled (e.g., Philippe $|^{couple}$ Chen). In this context, the meaning of a node by itself or even with respect to a single relationship may not be clear, hence a significant element of generation pipelines using knowledge graphs includes learning node representations (e.g., based on neighbors or the whole graph) [36]. Examples of this line of research involve knowledge graphs such as WebNLG or the AGENDA dataset [37, 38]. In contrast, the graphs of interest in this paper are *causal maps*.

Causal maps may erroneously be interpreted as a special case of knowledge graphs where edges only take two labels (positive or negative). While both knowledge graphs and causal maps are visually similar and both contain information, causal maps are explicitly unidirectional. When a knowledge graph shuffles its concepts and relationships, fine-tuned PLMs may bypass the graph's structure and refer back to its training data to avoid false information [39]. This becomes a conflict of interest for causal maps, as we *always* want the text to represent the input and not attempt to "fix itself", since the causal map should come from facilitators and experts. If the subgraph is resolved in the text generation, this gives the PLM a role as a facilitator to correct the human facilitator's work. Thus, we want the faithfulness to come from the facilitator rather than the PLM. Another way to say this is, we want to go from graph to text then back to graph seamlessly.

However, causal maps are distinct from knowledge graphs for two additional reasons. Nodes represent *measurable entities* (e.g., we know the meaning of having more or less physical activity) and do not usually need to be explained given the target audience of a map (e.g., stakeholders involved in obesity would understand 'food intake'). Consequently, less attention is devoted to learning node representations. In addition, causal maps serve to synthesize the domain for complex problems hence they often involve loops [3, 40, 41]. When such maps are used as precursors to System Dynamics models, they are known as causal *loop* diagrams. The linearization step must thus pay careful attention to addressing loops, for instance by elimination (i.e., the target model is a simplification of the input model) or by breaking the map into many smaller acyclic graphs (Figure 2).

2.2 Architectures for Graph-to-text Generation

Natural Language Generation has long been investigated, with early systems relying on hand-crafted rules or templates arranged into various modules such as discourse planning (e.g., a tree that specifies the ordering of paragraphs and sentences) or lexicalization (choosing specific words and phrases to express concepts and relations) [42]. Although these systems could create well-formed and accurate sentences, they were burdensome to design and implement. Early neural systems lessened the burden, but they were limited in scope and (unlike rule-based systems) they could create erroneous outputs such as nonsensical text, or text that is not faithful to the original input – this undesirable generation is known as *hallucination* [43]. This is obviously a problem for medical applications where a hallucinatory summary can be life-threatening. Hallucinations are also problematic in a participatory modelling context: participants may be dissatisfied with a model if they are given a wrong description, even if the model itself is correct. For these reasons, there are several measures about the quality of the output produced by neural systems. For instance, human readers can evaluate the output with respect to dimensions such as coverage (is the entire input described in the output?), faithfulness (does the output include constructs that were not part of, or run contrary to, the input?), and fluency (is the sentence well-formed?). In addition, there are several automatic metrics to compare the generated or 'hypothesis' text with a reference (e.g., BLEU, METEOR, and chrF [44]) and new ones continue to be proposed [45].

Early neural systems were succeeded by deep representational learning. Although different architectures are simultaneously examined in current research [46], we will briefly present the main ones from a historical perspective for the sake of simplicity. An early case of deep neural network was the Recurrent Neural Network (RNN) with an encoder-decoder structure [47]. Basic RNNs employ fixed-length vectors and struggle with long-term dependencies, hence some of the landmark studies used advanced RNNs such as Long Short-Term Memory (LSTM) networks to produce satisfying BLEU scores [48]. An LSTM is sequential: computing the current step relies on the output computed from the previous step. This prevents parallelism, thus creating a performance bottleneck as there are many parameters to train. Transformer models came next, using attention-based mechanisms, and can be trained in parallel. This performance improvement enabled the development of large-scale pre-trained language

models (PLMs) [49, 50] such as GPT-3, whose *temperature parameter* allows to "generate high-quality creative texts so that even humans cannot determine whether or not the texts are written by a human" [51]. Large-scale PLMs developed by OpenAI have attracted significant attraction via the success of ChatGPT, built on GPT-3. Most importantly, PLMs can leverage *transfer learning*: they learn universal language representations from a large volume of text and can reuse this knowledge for a specialized task. Instead of developing yet another model, many applications in NLG are thus shifting to the effective use of PLMs on specific tasks (e.g., summarization, translation, question answering).

In our prior work, we focused on using the large-scale PLM GPT-3 to generate explanatory sentences about a causal map [32]. Key steps of our process are summarized in Figure 2. Given a causal map, we performed linearization by first decomposing the graph into a set of overlapping communities (i.e., a node can exist in multiple communities) via the Core Expansion method. We then created a variant of a Breadth-First Search to break each community into small acyclic graphs with a specified depth limit, width limit, and maximum number of times that a node could be reused. This process was partially *destructive*, as a substantial loss of about 15% of nodes and 54% of edges happened in our case study. Subgraphs were converted to a description in the Resource Description Framework (RDF), Large Language Models (LLMs) such as GPT can process as input. Experiments produced satisfactory scores at the sentence-level for chrF and METEOR and high scores for faithfulness, fluency, and coverage.

3 Methods

3.1 Overview

Our methods are articulated along four main steps (Figure 3), which are explained in the following dedicated subsections. We start by comparing various community detection algorithms given a set criteria for our selection. This differs from our prior work, where only one community detection algorithm was used instead of optimizing its selection among state-of-the-art options, and where the algorithm lost both nodes and edges. Then, we implement a decomposition algorithm to traverse each community and extract an ordered list of subgraphs that corresponds to the RDF triplet as input for generating sentences. Next, we implement our revisions of pre-processing our data into prior work of generating sentences using a fine-tuned model. We have updated the base model for fine-tuning to GPT-3.5, expanded the original dataset to provide more instances for fine-tuning, and added a system prompt per GPT's fine-tuning format. Finally, we run experiments to find the best set of parameters for generating paragraphs. We compare various temperatures, bundle sizes (the number of sentences to be in one generated paragraph), and PLMs including base models of OpenAI's GPT (GPT-3.5-Turbo, GPT-4, and GPT-4-Turbo-Preview) [52, 53], Google's Gemini (gemini-1.0) [54], and Meta's Llama-2 (Llama-2-7b-chat-hf) [55]. For evaluation, we use automatic metrics to compare the faithfulness, fluency, and coverage of the generated paragraphs. We use a CDC report on suicide as our reference text, as the information directly reflects our causal map [56]. We provide a Flesch Reading

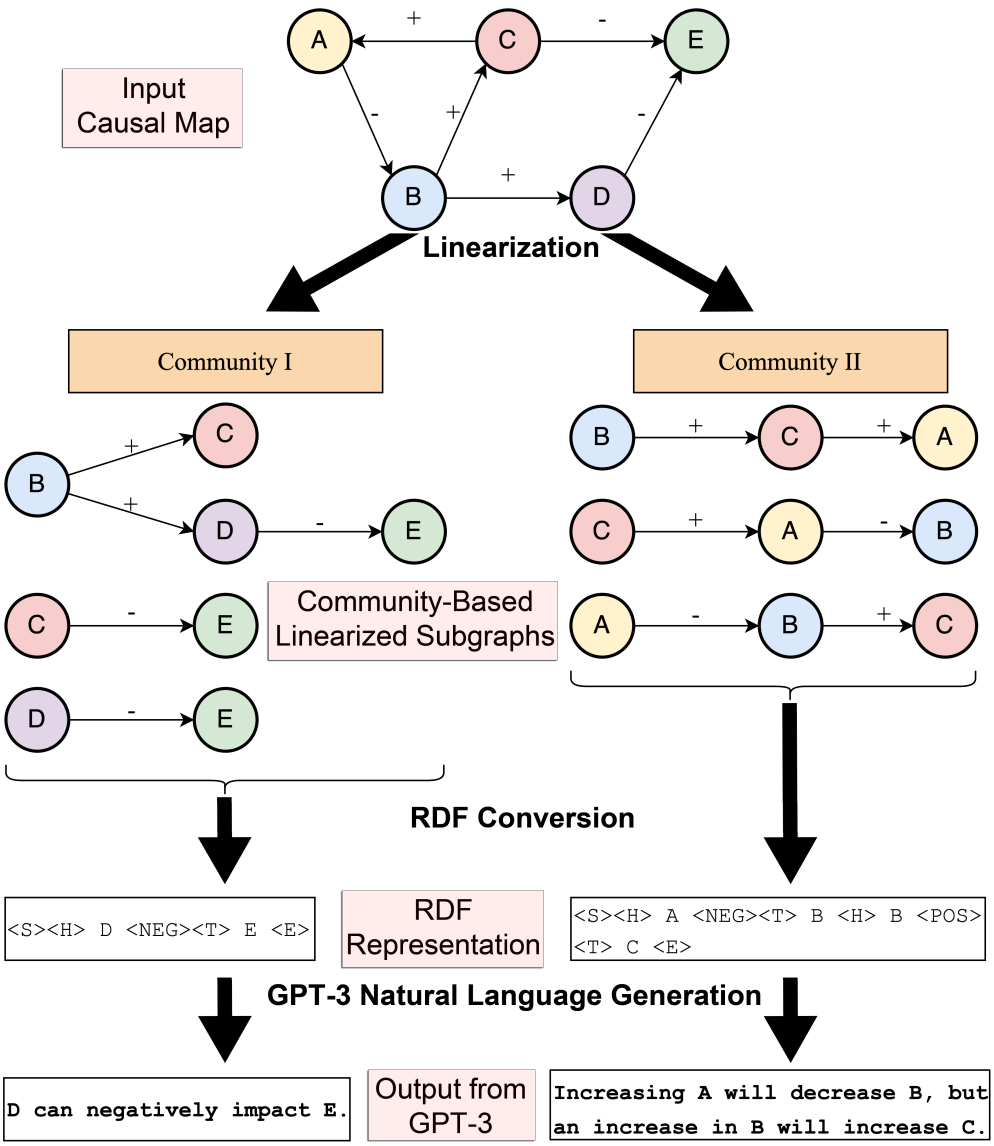

Fig. 2 Main steps of the process in our prototype [32].

Ease score for readability, and text evaluation metrics BLEU [57], ROUGE [58], and BERTScore [59] for similarity.

3.2 Decomposition Algorithms

We seek to decompose a given causal map into communities with a small amount of shared content (i.e., overlapping communities), since paragraphs need to be connected

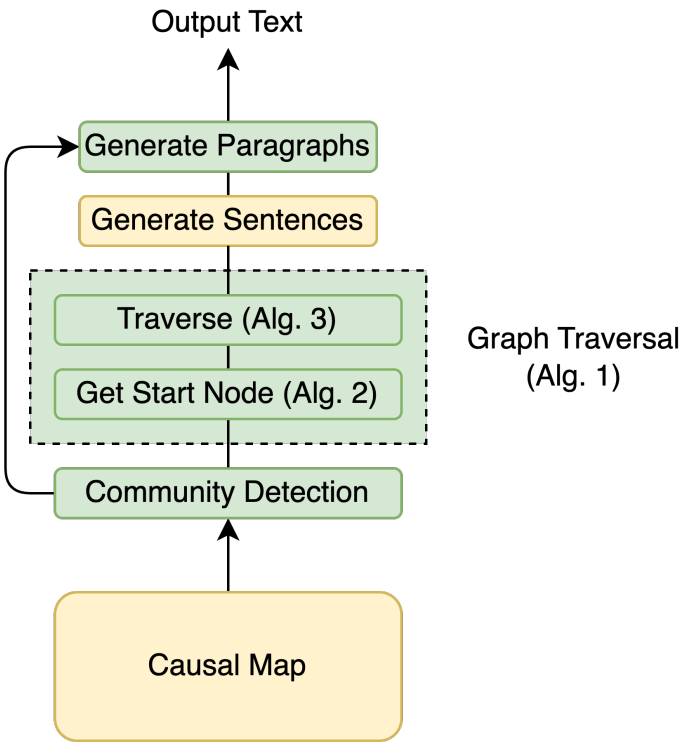

Fig. 3 Our main four steps to generating paragraphs about a causal map, emphasizing choosing the right community detection algorithm, utilizing graph traversal for each community, and using Shrestha *et al.*'s generation methods to generate paragraphs. Nodes in yellow define previous work [3, 32], while green nodes highlight our contributions.

by having some shared elements rather than jumping abruptly between topics. Consequently, we focus on community detection algorithms in static networks and evaluate them with respect to various requirements. One requirement is to *preserve* the graph, as the union of all communities should be as close as possible to the original graph. This criterion ensures that the paragraphs cover the content of the causal map. This criterion is measured through the number of missing nodes, with a target of 0 missing nodes. The other requirement is for our communities to have a high level of entropy, as observed by the distribution of the number of nodes in communities. This criterion seeks to avoid generating a few enormous paragraphs followed by masses of very short paragraphs. This is a common problem in community detections, as some algorithms lead to the "creation of monster communities" [60], typically with the first community being at least one order of magnitude larger than the rest. We compare 24 overlapping community detection algorithms, and calibrate 19 to find their best Newman-Girvan Modularity. Our full parameter search is demonstrated in Table 2. Once all algorithms are calibrated, we then compare each using Newman-Girvan Modularity again, along with coverage, entropy to measure distribution of communities,

and overlap ratio. Overlap ratio is calculated by observing how many nodes exist in more than one community.

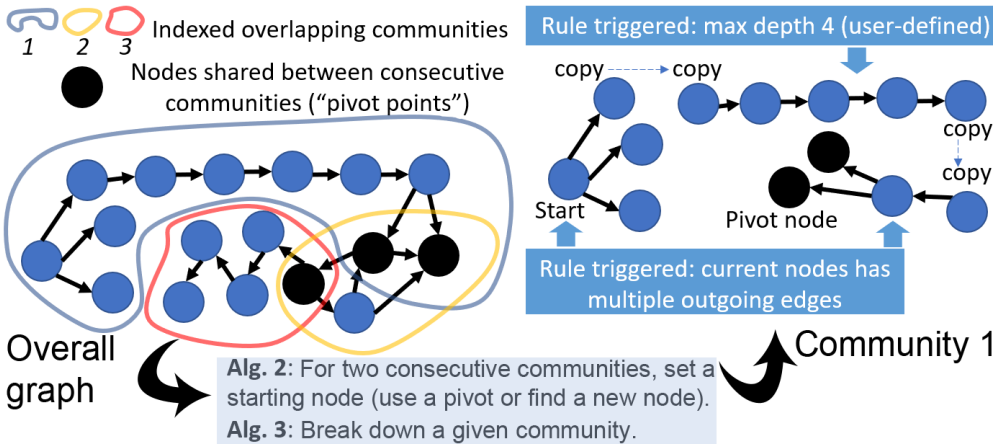

Fig. 4 Alg. 1 dispatches pairs of consecutive communities to Alg. 2 to find a starting node for each community, either using a pivot with the prior community (when available) or by finding the furthest node from the pivot with the next community. Alg. 3 then repeatedly uses a Breadth-First Search modified with two rules to subdivide each community without loss.

After selecting the best overlapping community detection algorithm given our criteria, we break each community into a set of components as illustrated in Figure 4. Note that the pseudocode provided here is complemented by our open-source implementation, hosted on a third-party repository at https://osf.io/98kmy/. In short, there are two levels of subdivisions: the communities, and their smaller components. Alg. 1 navigates through communities, identifying shared concepts or 'pivot nodes' (when they exist) that will be used to flow from the end of one paragraph to the beginning of the next one. Alg. 2 sets the starting node for the paragraph. When a paragraph flows from a prior one (i.e., they shared nodes), one of these pivot nodes is used as the start. For the first paragraph or for disjoint paragraphs, the starting node is identified as the furthest from the end of its paragraph. Alg. 3 repeatedly divides the current community into components. We prevent components from being exceedingly large by (i) capping long chains of causes and consequences $X_i \rightarrow \ldots \rightarrow X_k$ at a user-defined limit, and (ii) stopping when a node fans out to many consequences $X_i \rightarrow \{X_j, \ldots, X_k\}$. Alg. 3 ensures that no node or edge can be lost in the decomposition process.

3.3 RDF Conversion and Fine-tuning

Each of the components created through the process outlined in the previous subsection is transformed into the same RDF representation as in our prior work, as illustrated in Figure 2. In short, each pair of nodes $\mathbb{A}$ and $\mathbb{B}$ connected by an edge of type $\mathbb{T}$ will be turned into $< S >< H > \mathbb{A} < \mathbb{T} >< T > \mathbb{B} < E >$. S and E mark the start and end of a block, respectively; H and T represent the head and target node,

Algorithm 1: Initialize and dispatch

```
Input: List of n communities
Output: List of graph components
Function Init(graph with communities)
    Remove self-loops in the graph
    for every node v do
        /* A node is in a list of (at least one) communities */
        for i from 1 to n − 1 do
            If v is both in community i and i + 1, then flag v as pivot
    Call Alg. 2 on community 1 to get start node
    Pass start node to Alg. 3 to break 1st community
    for i from 2 to n do
        If communities i and i − 1 share a pivot, set it as start node
          Otherwise, find start node in i via Alg. 2
          Run Alg. 3 on i from the starting node
    return component list accumulated from Alg. 3
```

Algorithm 2: Select the starting node

```
Input: First two communities
Output: Starting node
Function GetStartNode(comm. j, comm. j + 1)
    If there are pivots between j and j + 1, use the first pivot as target node
        Otherwise, use source of 1st edge as target node
        Reverse direction of the graph
        Perform depth-first search from the target node to find the farthest
        node /* If there are many nodes with the same farthest
        distance, use 1st encountered */
    return the farthest node
```

respectively. For example, in the context of our application to Adverse Childhood Experiences (ACEs), one component of 9 edges can be turned into the following:

$< S >< H >$ Stigma of the ACE $< POS >< T >$ Cumulative trauma $< E > < S >< H >$ Fighting in the house $< POS >< T >$ Family risk factors $< E > < S >< H >$ Interpersonal problems $< POS >< T >$ ACEs $< E > < S >< H >$ ACEs of siblings $< POS >< T >$ ACEs $< E > < S >< H >$ Cultural norms that ACEs aren't problems $< POS >< T >$ ACEs $< E > < S>< H >$ Chronic exposure to unsafe messaging $< POS >< T >$ Feeling unsafe $< E > < S >< H >$ Healthy mentoring $< NEG >< T >$ ACEs $< E > < S >< H >$ Peer victimization $< POS >< T >$ ACEs $< E > < S >< H >$ Positive childhood experiences $< NEG >< T >$ ACEs $< E >$

Algorithm 3: Traverse and break community

```
Input: community i, start node, max depth
Output: list of components from community i
Function TraverseBreak(i, head, max depth)
    Queue of nodes Q ← [head]
    List of components L ← ∅
    while edges exist in community i do
        current component C ← ∅
        while Q is not empty do
            current depth d ← 1
                current component c ← ∅
                /* Starts a constrained breadth-first search */
            repeat
                current node v ← next in Q
                add to C all outgoing edges/nodes of v
                add to Q all outgoing nodes of v
                remove from i all outgoing edges of v
                increment depth d
            until outdegree(v) > d or d >= maxdepth;
            add component C to L
    return components L
```

We also manually created a dataset for fine-tuning, consisting of the desired text output for 697 edges. As fine-tuning was solely available with the GPT-3 model, we fine-tuned text-davinci-003 with 85% of our manually labeled data for training, 7.5% for validation, and and 7.5% for testing. These proportions are similar to other studies that fine-tune GPT models by devoting from 80% to 90% of the data to training [61–63] and using the rest either equally for validation and testing [62] or with a minimum of 5% devoted to testing [63]. All sentences were generated, one by one, through this fine-tuned model.

3.4 Paragraph Generation and Evaluation

To obtain a *baseline*, we output individual sentences in the paragraph exactly as they were generated in the previous subsection. Since our fine-tuned GPT model is only aware of one sentence at a time, it cannot smooth out a transition with the sentences immediately before or after. We then use our proposed *bundling approach*, which takes in a user-defined number of sentences (the 'bundle size') and assembles them in a paragraph. We used the prompt "You are a helpful assistant that combines a list of sentence into one paragraph in a way that a 15th grader could understand." For role-based models like GPT, this prompt is inserted as the system prompt for every paragraph. For OpenAI's GPT models, we experimented with GPT-3.5-Turbo, GPT-4, and GPT-4-Turbo-Preview [52, 53]. For Google's models, we utilized Gemini-1.0 [54]. For Meta, we used Llama-2-7b-chat-hf [55]. We thus have three parameters, whose values are optimized via a grid search by considering bundle sizes of {5, 10, 15,

20} and four values for temperature. PLMs have various ranges for temperature. GPT models are on a scale of 0 to 2, however temperatures higher than 1.0 tend to yield poor results in information and text generation. For consistency, we use the values {0.25, 0.5, 0.75, 1.0} for all models.

We use several evaluation metrics focusing on the faithfulness, fluency, and coverage of our text (as seen by Shrestha *et al.* in [32]). For faithfulness and coverage, we evaluate using n-gram comparison metrics BLEUand ROUGE [57, 58]. We also utilize BERTScore to evaluate the semantics of our text rather than n-gram similarity [59]. For fluency, we evaluate paragraphs with respect to their Flesch Reading Ease score, which scales from 0 to 100 [64]. Scores below 50 are difficult or "academic" (30–50), and scores below 30 are very difficult to read (0–30). This score has been used in several recent studies that generated text via GPT. These reports have primarily been in medical fields, and secondarily from fields such as education. These scores can range widely (Table 1), as they are the result of both variations in the use of technology as well as considerable differences in the inherent difficulty of texts originating from a given context. For example, we may expect a 4th grade explanation to have a higher readability level than surgical reports. It is thus important to contextualize these scores by also estimating the readability of *human* written texts for the same application.

Table 1 Studies evaluating the readability of GPT-generated paragraphs used different versions and for different applications.

Study	**Application**	**Technology**	**Average Flesch Reading Ease score (± standard deviation, when provided)**
[65]	radiology	ChatGPT 3.5*	83.5 ± 5.6
[66]	ophthalmology	ChatGPT 4.0	From 28.1±7.5 for epiretinal membrane to 34.4±7.7 for macular hole
[67]	dermatology	ChatGPT 4.0	42.67 (range 26.55-62.98)
[68]	drug-drug interactions	ChatGPT 3.5*	27.64±10.85 on asking whether 2 drugs can be taken together; 14.85±1.97 on why they shouldn't
[69]	4th grade history class	ChatGPT 3.5*	In English, 50 by default and averages 65 to 75 in child-friendly mode
[70]	college engineering class	GPT-3	43

**The specific version not stated by the authors and was inferred by publication dates.*

4 Results

To support replication and transparency efforts, our experimental evaluation used the code openly available[1] on our third-party repository at https://osf.io/98kmy/. As in our prior work [32], the input consists of a large causal map on Adverse Childhood Experiences (ACEs) and suicide in youth. This map was developed through a participatory modeling process detailed in [3], where fifteen subject matter experts provided

[1] To independently replicate our work, we recommend an environment with the same dependencies. The complete list of dependencies is provided on our repository. The main ones are: networkx 3.2.1 (to represent the causal map), cdlib 0.2.5 (community detection algorithms), openai 1.2.4 (for GPT), textstat 0.7.3 (for the readability score), and matplotlib 3.8.1 (to visualize results).

individual maps that were combined into an aggregate with 361 nodes and 946 edges. This map illustrates the problem addressed by our method: each expert provides a small map that is viewable and understandable by them, but participants cannot learn from each other given the complexity of the aggregated map. Since our prior work has demonstrated that *individual sentences* were satisfactory (in coverage, faithfulness, fluency), the emphasis of the present work is on orchestrating sentences in meaningful *paragraphs*. Our results thus focus on choosing the best overlapping community detection algorithm for our causal map and performing a quantitative analysis of our generated text.

4.1 Overlapping Community Detection

Table 2 shows our parameter search along with the best set of parameters based on Newman-Girvan Modularity. We utilize all ideal parameters alongside 5 more algorithms that do not require calibration, and evaluate based on Newman-Girvan modularity, entropy, coverage, and overlap (Table 3). We achieved the highest Newman-Girvan modularity score of 0.49 using an ensemble approach through EnD-NTM [71]. However, given the set of parameters, all communities were disjoint. The second-best score with 100% coverage and some overlap is Link Partitioning Around Medoids (LPAM) [72] with a modularity score of 0.32. Core expansion [73], Cluster-Overlap Newman Girvan Optimized (CONGO) [74], and Label Propagation Algorithm with Neighbor Node Influence (LPANNI) [75] achieved similar scores, all within 0.05 of the second-highest modularity score. However, Core Expansion and LPANNI are disjoint. CONGO and LPAM also have an equal overlap ratio, however CONGO has a significantly higher level of entropy. Given the significant jump in entropy, we sacrifice the slight decrease in modularity and utilize CONGO for processing our network traversal algorithms.

4.2 Evaluation of Paragraphs

To set expectations about the desired readability level, we tasked a team of human annotators with writing paragraphs in relation to the causal map. They manually wrote 27 paragraphs (provided as *human_sentences.txt* in https://osf.io/98kmy/), achieving a Flesch Reading Ease score of 53.92. For our automated system, we divided the causal map into overlapping communities using the CONGO overlapping community detection algorithm, calibrated for the highest modularity score.

Figure 5 displays Flesch Reading Ease heatmaps for all GPT models with varying temperatures and bundle sizes. Here we can see we achieve the best *readability* scores with GPT-3.5-Turbo with a temperature of 0.5 and bundle size of 5, and a worst score using GPT-4-Turbo-Preview with a temperature of 1.0 and bundle size of 20. These scores can also be seen in Figure 6, along with Gemini-1.0-Pro, Llama-2-7b-chat-hf, and our handwritten sentences.

Our average Flesch reading ease score was 36.371 ± 6.754. Llama-2 holds the top 6 scores with the best score of 46.88 using a temperature of 0.25 and bundle size of 15. Of those 6 results, none used a bundle size of 20. After the top Llama-2 scores, Gemini-1.0-Pro achieved a Flesch score of 45.86 with a temperature of 0.25 and bundle

Table 2 Grid search performed for multi-parameter algorithms. Most algorithms vary in search space due to their runtime, ranges, and number of parameters.

Algorithm	Parameter	Range	Best NG	Best Param
Angel [76]	Threshold Min. community size	0, 0.01, ..., 1.0 0, 1, ..., 49	0.028	0.73 3
CoAch [77]	Density threshold Affinity threshold Closeness	0, 0.0833, ..., 1.0 0, 0.0833, ..., 1.0 0, 0.0833, ..., 1.0	0.173	0 0.1667 0.0833
CONGA [78]	communities	10, 20, ..., 50	0.342	10
CONGO [74]	communities Height	10, 20, ..., 50 1, 2, 3	0.271	50 2
Core Expansion [73]	Tolerance	0.0001, 0.0011, ..., 1.0	0.305	0.0191
DEMON [79]	Epsilon Min. community size	0.1, 0.2, ..., 1.0 0, 1, ..., 49	0.220	0.5 1
Ego Networks	Level	0, 1, ..., 99	0.178	2
EnDNTM [71][1]	Epsilon	0.000, 0.001, ..., 100.0	0.490	0.00
IPCA [80]	t_in	0.00, 0.01, ..., 1.0	0.163	0.1
K-clique [81]	k	2, 3, ..., 9999	0.000	2
Lemon [82][2]	Min. community size Max.community size Expand step Substance dim. Walk steps Biased?	10, 20, 30 40, 50, ..., 100 1, 2, ..., 5 1, 2, ..., 5 1, 2, ..., 5 True, False	0.000	20 40 1 1 1 True
LFM [83]	Alpha	0.0000, 0.0042, ..., 0.8375	0.216	0.8249
LPAM [72]	k Threshold	25, 50 0.00010, 0.50005, 1.0	0.317	25 0.0001
LPANNI [75]	Threshold	0.0001, 0.0344, ..., 1.0000	0.275	0.0001
MULTICOM [84]	Seed node	0, 1, ..., 9	0.012	1
OSSE [85][2]	ninf? Expansion Alpha Delta	True, False 'ppr', 'vppr' 0.00, 0.01, ..., 1.0 0.0, 0.1, ..., 1.0	0.037	True 'ppr' 0.04 0
SLPA [86]	t r	10, 11, ..., 50 0.0, 0.1, 1.0	0.362	29 0.2
Walkscan [87]	Nb steps EPS Min. samples	1, 2, ..., 10 0.00, 0.01, ..., 1.0 1, 2, ..., 9	0.117	6 0.0101 6

[1]EnDNTM is an ensembling approach, and we used the default disjoint algorithms used by CDLib: Louvain [88], Label Propagation [89], and Walkscan [87].
[2]These algorithms utilized seed nodes, which we defaulted to "ACEs", "Suicide ideation", and "Suicide attempt".

size of 10. The bottom 10 Flesch reading ease scores are GPT-4 and GPT-4-Turbo-Preview, with the worst score being 21.13 with a temperature of 1.0 and bundle size of 20. Our original fine-tuned sentences spliced together achieved a score of 31.17.

Our BLEU scores are significantly low (maximum is 0.006), which is to be expected given that BLEU is designed for local text similarities rather than large documents [96]. GPT-4 and GPT-4-Turbo-Preview achieve ROUGE-1 scores over 0.3, with the highest score of 0.313. More precisely, the top 8 ROUGE-1 scores are all GPT-4 and GPT-4-Turbo-Preview models with a bundle size of 5. The next 6 scores are GPT-4 and

Table 3 Evaluation of calibrated algorithms, with the best scores displayed in bold. A negative NG modularity score is believed to be an implementation error, as scores less than 0 should not exist [95].

Algorithm	NG Modularity	Coverage	Entropy	Overlap
Angel [76]	0.01	0.47	3.70	0.03
ASLPAw [86]	0.04	0.99	0.69	0.00
CoAch [77]	0.18	0.86	5.31	0.16
CONGA [78]	0.21	**1.00**	0.94	0.02
CONGO* [74]	0.27	**1.00**	5.21	0.02
Core Expansion [73]	0.30	**1.00**	4.14	0.00
Demon [79]	0.23	**1.00**	4.41	0.04
EBGC [90]	0.05	**1.00**	7.79	0.00
Ego-networks	0.18	**1.00**	**8.00**	0.14
EnDNTM [71]	**0.49**	**1.00**	3.27	0.00
Graph Entropy [91]	0.08	**1.00**	7.72	0.00
IPCA [80]	0.16	0.88	6.69	0.07
k-clique [81]	0.00	**1.00**	0.00	0.00
LAIS2 [92]	-0.02	0.29	4.91	**0.48**
Lemon [82]	-0.05	0.09	0.00	0.00
LPAM* [72]	0.32	**1.00**	3.57	0.02
LPANNI [75]	0.27	**1.00**	2.10	0.00
LFM* [83]	0.22	**1.00**	1.75	0.02
MULTICOM [84]	-0.01	0.46	2.15	0.15
OSSE [85]	0.03	0.41	1.54	0.20
PercoMVC [93]	-0.02	0.17	1.89	0.05
SLPA [86]	0.08	**1.00**	0.91	0.01
UMSTMO [94]	0.04	**1.00**	7.29	0.00
Walkscan [87]	0.05	**1.00**	0.27	0.01

*Algorithm scores are the average of 200 runs.

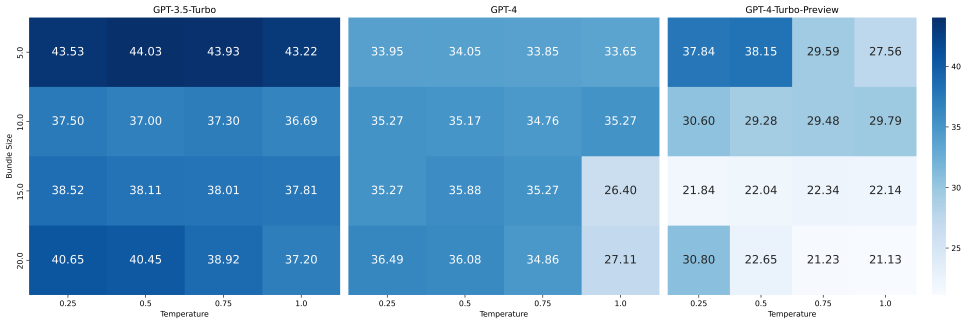

Fig. 5 Flesch Reading Ease Grid search on GPT models with varying temperature and bundle size.

GPT-4-Turbo-Preview models with a bundle size of 10, with the closest competitor being Gemini at 0.256 with temperature 1.0 and bundle size of 5. We see similar trends with ROUGE-2 and ROUGE-L, but our scores are significantly lower. Again, this is expected when we increase the n-gram length. BERTScore achieves significantly better results, with an average precision of 0.780 ± 0.009, recall of 0.728 ± 0.005, and F1-score of 0.753 ± 0.006.

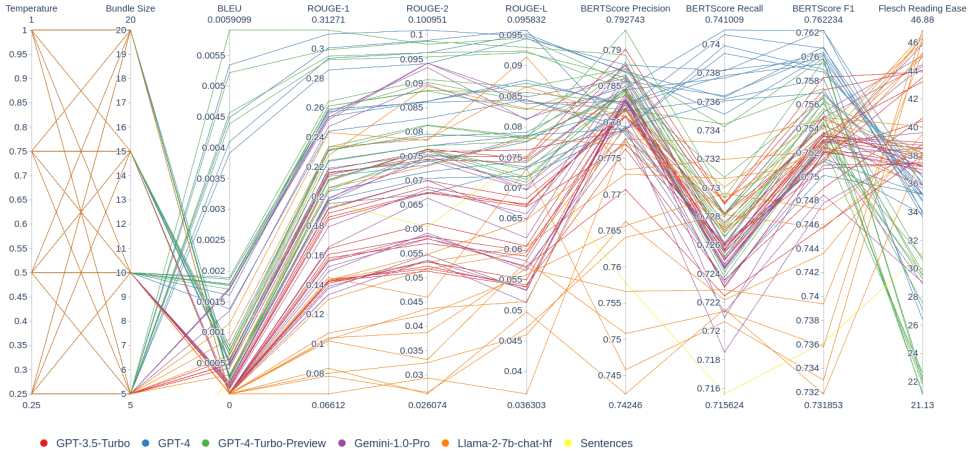

Fig. 6 Parallel coordinates plot for grid search on all 3 GPT models, Gemini-1.0, and Llama-2-7b-chat-hf with temperature and bundle size, along with the original sentences from fine-tuning.

Our qualitative inspection of paragraphs confirms that (i) they are written at a level that is accessible to a college-level reader, and (ii) they exhibit a satisfactory flow between consecutive paragraphs. As complete outputs are provided on our repository, we only provide three short examples for the end and beginning of consecutive paragraphs are follows.

> [...] Survivors of suicide attempts may have experienced more adverse childhood experiences and be more open to removing access to lethal means or participating in therapy.
>
> An increase in therapy can lead to disruptions and changes in parents' practices at home, while more therapy can reduce the chances of another attempt. [...]

> [...] By improving positive and hopeful thoughts, therapy can be more successful, leading to healthier relationships with children and increased parental involvement in their care. Feeling trapped and in pain can contribute to feelings of hopelessness, reduced energy and hopefulness, and an increased risk of suicide attempts, suicidal thoughts, and the traumatic experiences associated with adverse childhood experiences (ACEs).
>
> By encouraging children to seek help for mental health issues, they have a better chance of receiving the care they need and reducing negative experiences. Seeking help can lead to feeling more energetic, hopeful, and decrease mental health disorders. [...]

> [...] Increased isolation can reduce our access to life-saving treatment and resources, making us more vulnerable to suicide attempts and thoughts.

When there is less fear of death, it can make someone more likely to attempt suicide. [...]

In the first case, we note a clear shift from survivors of suicide attempts taking positive steps by participating in therapy, to what therapy can accomplish. In the second example, a feedback loop is described between paragraph transitions "By improving positive and hopeful thoughts, therapy can be more successful... Seeking help can lead to feeling more energetic, hopeful, and decrease mental health disorders." In the final instance, we see a transition using *suicide attempt* as a pivoting subject. Our qualitative assessment thus concludes that paragraphs have an appropriate transition, which validates our use of overlapping community detection algorithms.

5 Discussion

For quantitative evaluation, it is necessary to have human-written text as a reference point. While there is an abundance of large causal maps, none of them has so far been entirely explained through a *companion text document*. Instead of creating a handwritten textual representation of the causal map, we utilized the CDC's Suicide Prevention Resource [56]. While the general topics are the same, both resources (the causal map and report) were made different, and are not equivalent in content. This results in lower evaluation scores than desired. Generation of causal models would significantly benefit from a dataset with a causal map-list of texts pair that can be used as a benchmarking tool for a more generalized approach rather than one case study as we have demonstrated here, similar to the WebNLG dataset [97].

Our method requires *overlapping* communities, because paragraphs must have some shared elements in order to create the pivots that would be expected in fluent writing. However, this requirement does not necessarily mean that we are limited to using overlapping community detection algorithms. Indeed, our process could be revised to take in *disjoint* communities and expanding them to create overlaps, such as looking for edges where its head and target nodes reside in different communities instead of solely looking and nodes in multiple communities. Several solutions can already infer overlapping communities based on multiple disjoint community structures [98], such as the recent Expansion with Contraction method for Overlapping Community Detection (ECOCD) [99]. This could allow us to expand our search space and use algorithms for disjoint communities, which may enhance our results in regards to modularity and entropy. However, it adds one component to our process, hence we would have to optimize the disjoint community algorithm *and* the algorithm that creates overlaps.

When evaluating community detection algorithms, our parameter search space was selected based on the default values provided by CDLIb. Furthermore, we only used Newman-Girvan Modularity when selecting parameters instead of considering coverage, entropy, and overlap. While these metrics provide automatic scores for attributes e.g. structure (modularity), distribution (entropy) and coverage, we still lack a ground truth of manually-inspected communities to compare our community detection algorithms with. Even then, it can be difficult to identify which nodes may reside in

multiple communities, and to identify how many communities should exist. Experts are likely to make communities based on topics, while algorithms make communities based on structure. This study would significantly benefit from having a ground truth of communities, and if expanded to a dataset of causal maps and their respective communities, it may be feasible to create a general-case evaluation to determine an ideal community detection algorithm.

As discussed, the largest drawback of this paper is the use of a case study rather than a general case. If all aspects of data collection are implemented (datasets with aggregated causal maps, corresponding texts, ground truth communities), we could focus on improving the general case of causal maps to paragraphs, and thus allow this process to be more applicable to users. This would also invite other researchers to innovate the current process or invent their own methods to improve upon the current findings.

6 Conclusion

In this paper, we have proposed, implemented, and evaluated a new process to automatically turn a causal map (currently difficult to use for lay participants) into narratives (as a more accessible format). Our qualitative inspection of the generated paragraphs found that they had coherent topics and an appropriate flow from one paragraph to the next. For our case study of suicide in youth, we have identified a suitable overlapping community detection algorithm (CONGO), explained and applied our network traversal algorithms, generated ordered sentences using the output from network traversal and a fine-tuned GPT model, and performed a grid search on combining those sentences into paragraphs using various LLMs. We did not achieve our target score of 53.92, although our our text readability does lie in the academia reading level. We hope that our methods implore others to apply overlapping community detection algorithms in a similar manner, and provide a full analysis on algorithm calibration that meets the user's needs.

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Ethics declarations

Competing Interests

All authors declare that they have no conflicts of interest.

Ethical Compliance

This study does not involve any human participant or animal.

Data availability and access

Data is available on our third-party repository at https://osf.io/98kmy/.

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